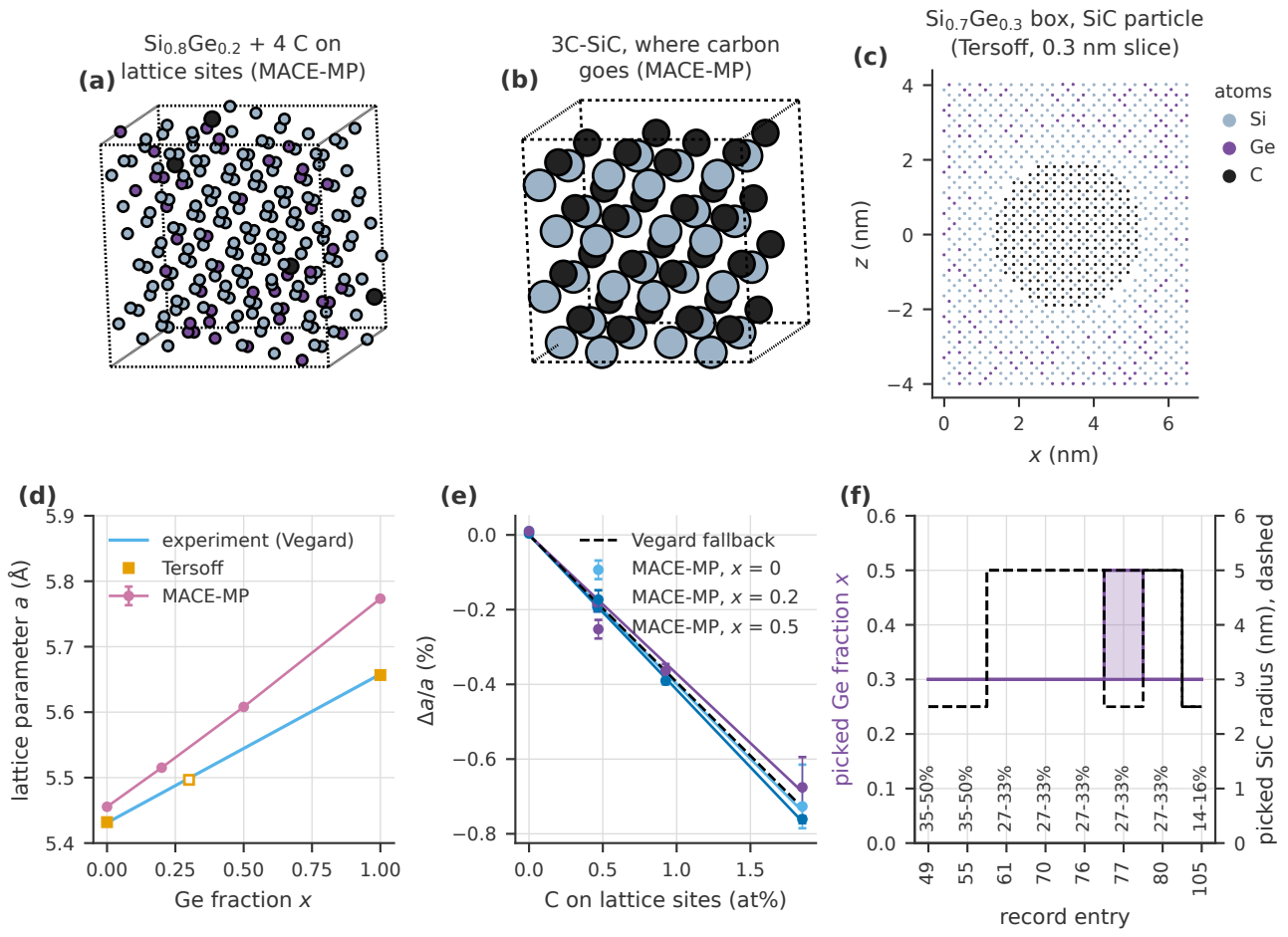

(a)
Si0.8Ge0.2 + 4 C on lattice sites (MACE-MP)
(b)
3C-SiC, where carbon goes (MACE-MP)
(c)
Si0.7Ge0.3 box, SiC particle (Tersoff, 0.3 nm slice)
atoms
Si
Ge
C
z (nm)
x (nm)
(d)
experiment (Vegard)
Tersoff
MACE-MP
lattice parameter a (Å)
Ge fraction x
(e)
Vegard fallback
MACE-MP, x = 0
MACE-MP, x = 0.2
MACE-MP, x = 0.5
Δa/a (%)
C on lattice sites (at%)
(f)
picked Ge fraction x
picked SiC radius (nm), dashed
35-50%
35-50%
27-33%
27-33%
27-33%
27-33%
27-33%
14-16%
record entry